

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA ANALYTICS [CSS 2103]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Non-programmable calculator permitted. Use the critical values supplied.

Section A: Multiple-choice questions (5 marks)

- 1) A manufacturer claims a population mean accuracy of 0.95. Which null hypothesis tests this claim? (1)
(A) $\mu = 0.95$
(B) $\bar{x} = 0.95$
(C) $\mu \neq 0.95$
(D) $\mu < 0.95$
- 2) A population has known standard deviation 0.02. For an independent sample of size 36, what is the standard error of the sample mean? (1)
(A) 0.12
(B) 0.02
(C) $0.02/6$
(D) $0.02/36$
- 3) With $\bar{x} = 0.947$, $\mu_0 = 0.95$, $\sigma = 0.02$ and $n = 36$, what is the z statistic? (1)
(A) 0.90
(B) -0.90
(C) -9.00
(D) 9.00
- 4) Observations range from 180 to 480. What is the width of each of three equal-width bins? (1)
(A) 60
(B) 150
(C) 300
(D) 100
- 5) Twelve observations are divided into three equal-frequency bins. How many observations are in each bin? (1)
(A) 3
(B) 4
(C) 6
(D) 12

Section B: Theory questions (25 marks)

- 6) (i) Find the standard deviation and skewness for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61. (5)
(ii) Give an example of the combining-variables concept in dataset preparation.
(iii) A CMO compares three new cola flavours using ratings from 12 customers on a 1-9 scale. Conduct the appropriate analysis. Ratings: Flavour A: 4, 2, 4, 2; Flavour B: 6, 4, 5, 5; Flavour C: 8, 6, 8, 6. Use population standard deviation and Pearson's second skewness coefficient. For (iii), assume independent normal samples with equal variances; use a one-way ANOVA at 5%, with critical $F(2,9) = 4.256$.
- 7) (i) Find the elements of a box plot for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61. (5)
(ii) Why is it sometimes necessary to create subsets from a large dataset before analysis?
(iii) Measure the strength of the relationship between the rankings assigned by two interviewers to candidates A-L. Interviewer 1: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. Interviewer 2: 1, 2, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11. For quartiles, use medians of the two halves. Use Spearman rank correlation in (iii).
- 8) A hospital laboratory claims that a new COVID testing kit correctly identifies infection 95% of the time. A researcher believes the true accuracy is less than 95%. A sample of 50 test results has mean accuracy 0.93; the known population standard deviation is 0.04. Test the claim at the 5% significance level. Critical z-value: -1.645. (3)
- 9) Test the effectiveness of vaccination in preventing smallpox at the 5% significance level. (5)
Vaccinated: attacked 31, not attacked 469, total 500.
Not vaccinated: attacked 185, not attacked 1315, total 1500.
Overall: attacked 216, not attacked 1784, total 2000. Use Pearson chi-square without continuity correction; critical value for 1 degree of freedom is 3.841. State association, not a causal conclusion.
- 10) Per-acre wheat production (tons) is recorded for three varieties and four fertilizers, with each combination replicated twice. Test whether variety differences and fertilizer differences are significant at the 5% level. Each pair below gives the two replicates, in fertilizer order F1, F2, F3, F4. (5)
Variety 1: (8,9), (9,8), (6,7), (10,11).
Variety 2: (7,6), (8,9), (5,4), (9,10).
Variety 3: (10,9), (11,12), (8,7), (12,13). Assume independent normal errors with common variance. Include the interaction term. Use critical F values: variety (2,12): 3.885; fertilizer (3,12): 3.490.
- 11) A data scientist is analysing the following cholesterol level (mg/dL) readings of 12 patients: [145, 155, 160, 162, 165, 172, 180, 190, 200, 220, 240, 260]. Apply equal-width and equal-frequency binning with 3 bins. Using histogram analysis, explain which binning approach (width-based or frequency-based) better represents the data distribution and why. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA STRUCTURES [CSS 2101]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2; module 3 restricted to stacks. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which data structure directly supports undoing the most recent action first? (1)
(A) Queue
(B) Stack
(C) Sorted array
(D) Adjacency matrix
- 2) Edits A, B and C are pushed in that order onto an undo stack. Which edit does the first undo remove? (1)
(A) A
(B) B
(C) All three
(D) C
- 3) After edits A, B and C, two undo operations transfer actions to a redo stack. What is the redo stack from bottom to top? (1)
(A) C, B
(B) B, C
(C) A, B
(D) A, C
- 4) The redo stack contains C, B from bottom to top. Which edit is reapplied by one redo operation? (1)
(A) A
(B) C
(C) B
(D) Both B and C
- 5) In a conventional two-stack editor, what happens to the redo stack when a new edit is made after an undo? (1)
(A) It is reversed
(B) It is cleared
(C) It is unchanged
(D) It becomes the undo stack

Section B: Theory questions (25 marks)

- 6) Write a C program to read and display employee details including the date of joining using nested structure. Consider date of joining as nested structure. (2)
- 7) Assume that you are storing marks of students in 3 subjects. Write a C program that: i. Takes the number of students as input. ii. Dynamically allocates a 2D array using pointers. iii. Accepts marks for each subject. Calculates and prints the average marks of each student using only pointer arithmetic. (2)
- 8) Write a C program to dynamically allocate memory for 2D string arrays (array of strings) and perform concatenation using pointers. (2)
- 9) A music streaming service keeps track of songs in a playlist using a singly linked list. Each node stores the following details: SongID (integer), Title (string), Duration (in seconds). Write a C program to perform the following operations: Insert songs into the playlist in the order they are added. Remove all songs whose duration is less than a user-specified value (Use head pointer for performing this operation). Display the updated playlist. Include a main() function to demonstrate all functionalities. (5)
- 10) A mobile app named "MIT-BLR Connect" maintains contact details of alumni in three different modules: i) Add Recent Contacts Module: Displays recently interacted alumni in order of contact - new ones added at the beginning. ii) Traverse Module: Allows viewing alumni details in forward direction. iii) Develop a module to print the data in reverse order. Develop C functions considering singly linked list as the suitable data structure (5)
- 11) Write a function to convert an infix expression to postfix. Include the following functions as per the prototype. i) int InStackPrecedence(char op) ; to return the precedence of operator within the stack ii) int IncomingPrecedence(char op); to return the precedence of the incoming operator. iii) void Infix_Postfix(char infix[], char postfix[]); to convert given infix expression to postfix. Main function not required. Note: The infix expression may include ONLY the following operators (listed in the order of their precedence): ; ^ (exponentiation) → right associative ; * (multiplication), / (division) → left associative ; + (addition), - (subtraction) → left associative (5)
- 12) Write a C function bool isDuplicate(char s[]) that detects duplicate parentheses in a given expression using a stack. (2)
- 13) Using the following nested calls, justify why a stack is preferred over a queue in handling function calls. Explain the role of activation records (stack frames).
void fun1() { printf("Inside fun1\n"); }
void fun2() { fun1(); printf("Inside fun2\n"); }
int main() { fun2(); printf("Inside main\n"); } (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA COMMUNICATION AND COMPUTER NETWORKS [CSS 2102]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) A 900-byte payload passes through five layers, each adding a 20-byte header. With no trailers, what is the transmitted size? (1)
(A) 920 bytes
(B) 1000 bytes
(C) 1020 bytes
(D) 1100 bytes
- 2) A transmitted unit contains 900 payload bytes and 100 header bytes. What percentage of the total is headers? (1)
(A) 9%
(B) 11.11%
(C) 90%
(D) 10%
- 3) What is the polynomial degree of CRC generator 10011? (1)
(A) 3
(B) 5
(C) 4
(D) 2
- 4) How many zeros are appended to the data before CRC division using generator 10011? (1)
(A) 4
(B) 5
(C) 3
(D) 1
- 5) Which operation performs subtraction in CRC modulo-2 division? (1)
(A) AND
(B) XOR
(C) OR
(D) Decimal subtraction

Section B: Theory questions (25 marks)

- 6) Compare Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) in terms of how each modifies the carrier signal, their bandwidth efficiency, and susceptibility to noise. Analyse which technique is most suitable for transmission over a noisy telephone channel and justify your choice. (6)

- 7) Define data rate and bandwidth. A square wave is approximated by (6)

$$s(t) = \frac{4}{\pi} \left[\sin(2\pi \cdot 2 \cdot 10^6 t) + \frac{1}{3} \sin(2\pi \cdot 6 \cdot 10^6 t) + \frac{1}{5} \sin(2\pi \cdot 10 \cdot 10^6 t) \right].$$

Assume two bits per fundamental cycle and bandwidth equal to highest minus lowest retained frequency. Calculate the data rate and bandwidth. Triple all frequencies and find the new values. Explain how bandwidth affects data rate.

- 8) For the given binary data 101000101010, draw the following line coding scheme. (Assume previous pulse was negative) i. Differential Manchester ii. Bipolar AMI For Differential Manchester, use a boundary transition for 0 and none for 1; initial level is negative. A mid-bit transition always occurs. (2)
- 9) For the given binary data 111000101010, draw the following line coding scheme. (Assume previous pulse was negative) i) Manchester ii) Bipolar AMI Manchester convention: 1 is low-to-high and 0 is high-to-low at mid-bit. (2)
- 10) A company network consists of three departments connected as follows: Sales and HR are connected through a hub, HR and IT are connected through a switch, and the switch is connected to a router for internet access. An employee in Sales sends a file to an employee in IT. Analyse how the hub, switch, and router handle this transmission. (2)
- 11) A sender uses a window of 3 to transmit frames 0, 1 and 2. Frame 1 is lost, while 0 and 2 arrive. ACKs are immediate and never lost; no new frames are available. In go-back-N, discard out-of-order frames and retransmit all outstanding frames at timeout. In selective repeat, buffer out-of-order frames and retransmit only the unacknowledged frame. For each protocol, list retransmissions and calculate useful frames divided by total data transmissions. (3)
- 12) Calculate the total time required to transmit 1,000,000 bits using the Stop-and-Wait ARQ protocol when each frame carries 2000 bits, the sender and receiver are 5000 km apart, and the propagation speed is 2×10^8 m/s. Ignore transmission, waiting, and processing delays, and assume no frames are lost or damaged Count one full round-trip per frame, including the final ACK. (2)
- 13) An Ethernet segment has transmission speed 10^8 bits/second and maximum length 1500 metres. If propagation speed is 3×10^8 metres/second, calculate the minimum frame size in bits required for collision detection. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-4; module 5 through memory operations. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) What is the range of 5-bit two's-complement integers? (1)
(A) -15 to 15
(B) -16 to 15
(C) -16 to 16
(D) 0 to 31
- 2) What is the 5-bit two's-complement representation of -9 ? (1)
(A) 01001
(B) 11001
(C) 10110
(D) 10111
- 3) Which addition overflows the 5-bit two's-complement range? (1)
(A) $-9 + 5$
(B) $6 - 13$
(C) $11 + 7$
(D) $-4 + 3$
- 4) Which arrays are programmable in a programmable logic array (PLA)? (1)
(A) Both AND and OR
(B) Only AND
(C) Only OR
(D) Neither
- 5) A PLA implements $F_1 = AB + AC$ and $F_2 = AB + BC$. Which product term can be shared? (1)
(A) AC
(B) AB
(C) BC
(D) ABC

Section B: Theory questions (25 marks)

- 6) Simplify $F(A, B, C, D) = \Pi M(0, 3, 5, 6, 8, 10)$, with don't-care terms $d(2, 13, 14)$, and implement it using NOR gates only. Take A as the most significant bit. (3)
- 7) Design a 3-bit magnitude comparator for unsigned binary numbers $A = a_2a_1a_0$ and $B = b_2b_1b_0$. Derive the Boolean expressions for $A > B$, $A = B$ and $A < B$, and draw the equivalent logic circuit. (3)
- 8) Implement a three-input XOR gate using 4-to-1 multiplexer with a neat diagram. (3)
- 9) Design a sequential circuit with two JK flip-flops A and B and two inputs E and F. If $E = 0$, the circuit remains in the same state regardless of the value of F. When $E = 1$ and $F = 1$, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When $E = 1$ and $F = 0$, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats. (5)
- 10) Design a 3-bit synchronous binary up-counter using T flip-flops. Draw the logic diagram. (3)
- 11) Illustrate a 4-bit asynchronous ripple counter with a neat diagram using JK Flip flops and explain the working of the same. (3)
- 12) Write behavioral Verilog code that represents an eight-bit ring counter. (2)
- 13) Explain the concept of memory addresses in an Instruction Set Architecture. Also, discuss the difference between byte addressing and word addressing. (3)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DISCRETE MATHEMATICAL STRUCTURES [MAT 2101]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

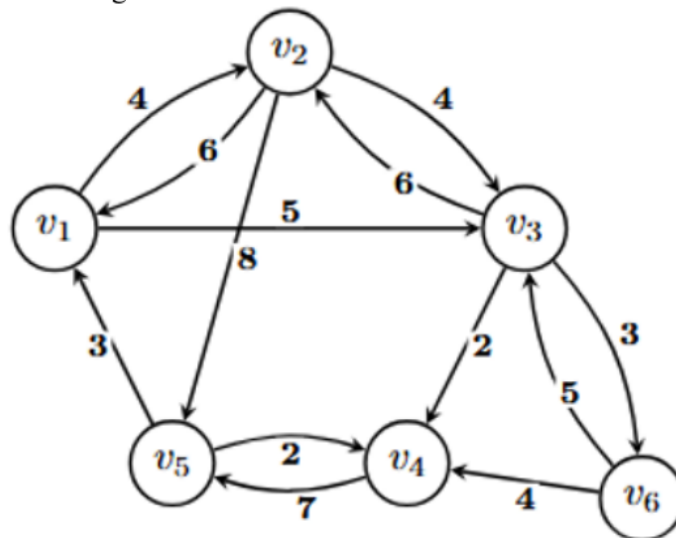
Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) Which pair of conditions defines a complement b of a in a Boolean algebra? (1)
(A) $a \vee b = 0, a \wedge b = 1$
(B) $a \vee b = 1, a \wedge b = 0$
(C) $a \vee b = a, a \wedge b = b$
(D) $a = b$
- 2) How many complements does each element of a Boolean algebra have? (1)
(A) None
(B) At least two
(C) Infinitely many
(D) Exactly one
- 3) Which expression equals $a \vee (\neg a \wedge b)$? (1)
(A) $a \vee b$
(B) $a \wedge b$
(C) $\neg a \vee b$
(D) a
- 4) Which expression is in disjunctive normal form (a disjunction of conjunctions of literals)? (1)
(A) $(x \vee y) \wedge z$
(B) $\neg(x \vee y)$
(C) $(x \wedge y) \vee (\neg x \wedge z)$
(D) $(x \vee y) \wedge (\neg x \vee z)$
- 5) Which expression is in conjunctive normal form (a conjunction of disjunctions of literals)? (1)
(A) $(x \wedge y) \vee z$
(B) $(x \vee y) \wedge (\neg x \vee z)$
(C) $\neg(x \wedge y)$
(D) $(x \wedge y) \vee (\neg x \wedge z)$

Section B: Theory questions (25 marks)

- 6) Draw the Hasse diagram for $A = \{1, 2, 3, 12, 18, 36\}$, ordered by divisibility: a is related to b iff a divides b . Is it a lattice? Justify. List all antichains of length more than one. (5)
- 7) How many ways are there to select a total of 15 items from three types, subject to these limits? (4)
Type A: 1 to 4 items; Type B: 3 to 7 items; Type C: 5 to 10 items.
- 8) Show that the number of partitions of n with no part smaller than 3 equals the number of partitions of $n+3$ whose three largest parts are consecutive integers. (3)
- 9) Find the 75th permutation of the five marks 0, 1, 2, 3, 4 in Fike's order. (3)
- 10) Show that if G is a regular self-complementary graph of order n , then $n = 4t + 1$ for some integer t . (3)
- 11) If G has p vertices and minimum degree $\delta(G) \geq (p - 1)/2$, prove that G is connected. (3)
- 12) Use Dijkstra's algorithm to find the shortest paths from v_1 to every other vertex in the directed network below. Show the steps. (4)



End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA ANALYTICS [CSS 2103]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

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Non-programmable calculator permitted. Use the critical values supplied.

Section A: Multiple-choice questions (5 marks)

- 1) What is the mean of 4, 6, 6, 8, 10, 14? (1)
(A) 6
(B) 7
(C) 8
(D) 10
- 2) What is the median of 4, 6, 6, 8, 10, 14? (1)
(A) 6
(B) 7
(C) 8
(D) 9
- 3) The six observations 4, 6, 6, 8, 10, 14 have mean 8 and squared-deviation sum 64. What is their population variance? (1)
(A) $64/6$
(B) $64/5$
(C) $\sqrt{64/6}$
(D) 8
- 4) A distribution has mean 8, median 7 and positive standard deviation. What is the sign of Pearson's second skewness coefficient? (1)
(A) Zero
(B) Negative
(C) Undefined
(D) Positive
- 5) Which measure of central tendency is generally less affected by an extreme observation? (1)
(A) Arithmetic mean
(B) Median
(C) Midrange
(D) All equally affected

Section B: Theory questions (25 marks)

- 6) Find the five-number summary and interquartile range of 3, 5, 7, 8, 9, 11, 13, 16, 30. Exclude the median when forming the lower and upper halves. Use the 1.5 *IQR* rule to identify outliers and sketch a box plot. (3)
- 7) A table contains customer ID, age, monthly income and annual income. Give one suitable treatment for missing ages and explain how you would detect inconsistency between the two income fields. (2)
- 8) A filling machine is claimed to dispense a mean of 500 ml. A random sample of 36 bottles has mean 496 ml. The population standard deviation is known to be 12 ml. Perform a two-sided *z*-test at the 5% level. State the hypotheses, statistic, decision and conclusion. Use critical values ± 1.96 . (5)
- 9) Two judges rank six projects. Judge A assigns ranks (1, 2, 3, 4, 5, 6) and judge B assigns (2, 1, 4, 3, 6, 5). Calculate Spearman's rank correlation coefficient and interpret the result. (3)
- 10) Partition 10, 12, 15, 18, 20, 22, 25, 28, 30, 32, 35, 40 into three equal-frequency bins. Replace each observation by its bin mean and report the three resulting bin means. (2)
- 11) Test independence between training and passing an assessment using Pearson's chi-square test at 5%. Of 60 trained students, 48 pass and 12 fail; of 40 untrained students, 22 pass and 18 fail. Show expected frequencies and the statistic. Use $\chi^2_{0.05,1} = 3.841$; do not apply a continuity correction. (5)
- 12) Three independent normal populations have equal variances. Samples are A: 4, 5, 6, B: 7, 8, 9, C: 10, 11, 12. The between-group sum of squares is 54 and within-group sum of squares is 6. Complete the one-way ANOVA degrees of freedom, mean squares and *F* statistic. Use $F_{0.05;(2,6)} = 5.143$ and state your conclusion. (3)
- 13) Distinguish a Type I error from a Type II error in the context of the filling-machine test in Question 8. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA STRUCTURES [CSS 2101]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

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Section A: Multiple-choice questions (5 marks)

- 1) Which C keyword defines a structure type? (1)
(A) `class`
(B) `union`
(C) `struct`
(D) `enum`
- 2) Object `s` has a nested structure member `admitted` with an integer member `year`. How do you access that `year`? (1)
(A) `s.admitted.year`
(B) `s->admitted.year`
(C) `s.year.admitted`
(D) `s::admitted::year`
- 3) How many non-null characters can a null-terminated string in `char name[40]` hold? (1)
(A) 40
(B) 41
(C) 38
(D) 39
- 4) Which `scanf` format limits a single word to fit in `char name[40]`, including its null terminator? (1)
(A) `%40s`
(B) `%39s`
(C) `%s`
(D) `%41s`
- 5) If `p` points to a structure with integer member `roll`, which expression accesses that member? (1)
(A) `p.roll`
(B) `*p.roll`
(C) `p->roll`
(D) `p::roll`

Section B: Theory questions (25 marks)

- 6) Write a C function that dynamically allocates a contiguous $r \times c$ integer matrix and returns its base pointer, or NULL on failure. Reject nonpositive dimensions. Show how element (i, j) is accessed using pointer arithmetic. Assume the requested size fits in `size_t`. (3)
- 7) Explain a dangling pointer and a memory leak using one short C example of each. State how each error can be prevented. (2)
- 8) A singly linked list stores integer values. Define its node structure and write `void deleteAll(Node **head, int key)` to delete every node whose value equals `key`. Handle an empty list, consecutive matches and matches at the head. Release deleted nodes. (5)
- 9) Write a C function to reverse a singly linked list in place using three pointers. Return the new head and state its time and auxiliary-space complexity. (3)
- 10) A list initially contains $10 \rightarrow 20 \rightarrow 30$. Insert 5 at the head, insert 25 after 20, then delete the tail. Draw the final list and state how many next links must be traversed to locate its tail from the head. (2)
- 11) Convert $A+B*(C-D)^E^F$ to postfix using a stack. Exponentiation has highest precedence and is right-associative. Show the operator stack and output after each input token, then give the final postfix expression. (5)
- 12) Write a C function to check whether a string containing only `(, ), [, and ]` is balanced. Use an array stack of capacity 100 and assume the input has at most 100 characters. Return 1 for balanced and 0 otherwise. (3)
- 13) An editor uses undo and redo stacks. Starting with empty stacks, perform edits A, B and C, then undo twice and redo once. List both stacks from bottom to top. State what happens to the redo stack when a new edit D is made. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA COMMUNICATION AND COMPUTER NETWORKS [CSS 2102]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-2. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) A noiseless channel has bandwidth 3 kHz and four signal levels. What is its Nyquist maximum bit rate? (1)
(A) 3 kbps
(B) 6 kbps
(C) 12 kbps
(D) 24 kbps
- 2) How many bits can one of four distinct signal levels represent? (1)
(A) 1
(B) 2
(C) 4
(D) 8
- 3) A channel has bandwidth 3 kHz and linear signal-to-noise ratio 31. What is its Shannon capacity? (1)
(A) 15 kbps
(B) 12 kbps
(C) 93 kbps
(D) 30 kbps
- 4) For the same channel and signalling scheme, Nyquist gives 12 kbps and Shannon gives 15 kbps. Which is the tighter upper bound? (1)
(A) 15 kbps
(B) 27 kbps
(C) 3 kbps
(D) 12 kbps
- 5) Under the noiseless Nyquist model, what happens to the maximum bit rate when bandwidth doubles and the number of levels stays fixed? (1)
(A) It halves
(B) It stays fixed
(C) It doubles
(D) It quadruples

Section B: Theory questions (25 marks)

- 6) Compare ASK, binary FSK and BPSK in terms of the carrier parameter changed and sensitivity to amplitude noise. State one reason to prefer BPSK over ASK in a channel with amplitude disturbances. (3)
- 7) An application generates 900 payload bytes. Each of five layers adds a 20-byte header. Calculate the total transmitted size and the percentage occupied by headers, ignoring trailers. (2)
- 8) Encode data 1101011011 using CRC generator 10011. Show modulo-2 division, the four-bit remainder and the complete transmitted codeword. State the receiver's check for an error-free received word. (5)
- 9) Draw Manchester and bipolar AMI waveforms for 10110010. For Manchester, use low-to-high at mid-bit for 1 and high-to-low for 0. For AMI, assume the last nonzero pulse before this sequence was negative. Label all bit boundaries. (3)
- 10) Explain the different actions of a hub and a learning switch when a frame arrives. For the switch, distinguish known and unknown destination MAC addresses. (2)
- 11) A stop-and-wait link sends 1000-byte data frames at 1 Mbps. One-way propagation delay is 10 ms. ACK transmission and processing times are negligible; no frames are lost. Calculate frame transmission time, time per successful cycle, utilization and useful throughput. Ignore headers. (5)
- 12) A selective-repeat sender sends frames 0, 1, 2 and 3 with window size 4. Frame 1 is lost; frames 0, 2 and 3 arrive and their individual ACKs return before timeout. No new frames are available. State which frames are buffered, which frame is retransmitted and the final in-order delivery sequence. Compare with go-back-N under the same loss, assuming it discards out-of-order frames. (3)
- 13) A shared half-duplex Ethernet link operates at 100 Mbps. The maximum one-way propagation delay is $5\ \mu\text{s}$. Ignoring other delays, calculate the minimum frame length required for collision detection in bits and bytes. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-4; module 5 through memory operations. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) How many cells are in a four-variable Karnaugh map? (1)
(A) 4
(B) 8
(C) 16
(D) 32
- 2) How many input variables differ between two adjacent Karnaugh-map cells? (1)
(A) 1
(B) 2
(C) 3
(D) 4
- 3) Which is a valid group size when minimizing a sum-of-products expression on a Karnaugh map? (1)
(A) 3
(B) 5
(C) 6
(D) 8
- 4) For $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$, with A most significant, which expression is equivalent? (1)
(A) $B \oplus D$
(B) $\neg B \neg D + BD$
(C) $B + D$
(D) BD
- 5) How can a two-input NAND gate act as an inverter for input X? (1)
(A) Tie both inputs to 0
(B) Leave one input open
(C) Tie both inputs to X
(D) Tie one input to 0

Section B: Theory questions (25 marks)

- 6) Using 5-bit two's complement, evaluate (i) $11 + 7$, (ii) $-9 + 5$, and (iii) $6 - 13$. Give the binary result, its signed interpretation and whether signed overflow occurs in each case. (3)
- 7) Implement $F(A, B, C) = A \oplus B \oplus C$ using a 4-to-1 multiplexer. Choose A and B as select inputs and specify I_0, I_1, I_2, I_3 ; an inverter is available. (2)
- 8) Design a synchronous two-bit counter with T flip-flops and enable input E. For $E = 1$, the state sequence is $00 \rightarrow 01 \rightarrow 10 \rightarrow 11 \rightarrow 00$; for $E = 0$, the state holds. Draw the transition table, derive both T-input equations and draw the circuit. (5)
- 9) Write synthesizable Verilog for a four-bit ring counter with positive-edge clock and active-high synchronous reset. Reset must load 0001; subsequent clocks rotate the one toward the next more significant bit, wrapping from bit 3 to bit 0. (3)
- 10) Explain why an asynchronous ripple counter can briefly produce incorrect decoded outputs. Contrast its clock connection with that of a synchronous counter. (2)
- 11) Design a two-bit unsigned magnitude comparator for $A = a_1a_0$ and $B = b_1b_0$. Derive Boolean expressions for $A > B$, $A = B$ and $A < B$. Draw a gate-level diagram and verify the outputs for $A = 10_2$, $B = 01_2$. (5)
- 12) A byte-addressable memory stores 32-bit words. How many words fit in 4 KiB? How many address bits select individual bytes? If a word starts at hexadecimal address $0x1000$, list the byte addresses it occupies and the starting address of the next word. (3)
- 13) Describe a memory-read operation using the memory address register (MAR), memory data register (MDR) and read control signal. State what changes for a memory-write operation. (2)

End of question paper

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DISCRETE MATHEMATICAL STRUCTURES [MAT 2101]
COMPUTER SCIENCE AND ENGINEERING
SET B: NEWLY AUTHORED QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Section A: Multiple-choice questions (5 marks)

- 1) In the divisibility order on $D_{12} = \{1, 2, 3, 4, 6, 12\}$, what is the least element? (1)
(A) 0
(B) 2
(C) 1
(D) 12
- 2) In the divisibility order on $D_{12} = \{1, 2, 3, 4, 6, 12\}$, what is the greatest element? (1)
(A) 1
(B) 12
(C) 6
(D) 4
- 3) In the divisor lattice D_{12} , what is $4 \wedge 6$? (1)
(A) 2
(B) 4
(C) 6
(D) 12
- 4) In the divisor lattice D_{12} , what is $4 \vee 6$? (1)
(A) 2
(B) 4
(C) 6
(D) 12
- 5) Which is a complement of 2 in D_{12} , with meet given by gcd and join by lcm? (1)
(A) 3
(B) 6
(C) No element
(D) 12

Section B: Theory questions (25 marks)

- 6) For $F(x, y, z) = (x \wedge \neg y) \vee z$, construct the truth table and write the canonical disjunctive normal form and canonical conjunctive normal form. Use variable order (x, y, z) . (3)
- 7) Prove the Boolean identity $a \vee (\neg a \wedge b) = a \vee b$ using distributive and complement laws. Name the laws used. (2)
- 8) Count the integer solutions of $x + y + z = 12$ subject to $1 \leq x \leq 5$, $2 \leq y \leq 6$ and $z \geq 0$. Transform to nonnegative variables and use inclusion-exclusion. Show each excluded case. (5)
- 9) Find the 42nd lexicographic permutation of the symbols 0, 1, 2, 3, 4, with 01234 counted as the first. Show the factorial-block choices. (3)
- 10) How many distinct arrangements of the letters of BALLOON are possible? How many have the two Ls adjacent? Explain how repeated letters are handled. (2)
- 11) Use Dijkstra's algorithm from A in an undirected weighted graph with vertices A, B, C, D, E and edges AB:4, AC:1, BC:2, BD:1, CD:5, CE:8, DE:3. Show the tentative-distance table after settling each vertex. Give a shortest path and distance from A to every other vertex. (5)
- 12) Prove that a simple graph with $n \geq 2$ vertices and minimum degree at least $(n - 1)/2$ is connected. Hint: bound the size of each component if the graph were disconnected. (3)
- 13) A simple graph is regular and self-complementary on n vertices. Use degree counting to show its degree is $(n - 1)/2$ and then show that $n \equiv 1 \pmod{4}$. (2)

End of question paper